

In [2]:

```
import pandas as pd

df1 = pd.DataFrame({"a": [34, 56, 45, 67], "b": [32, 67, 96, 33]})
df2 = pd.DataFrame({"c": [89, 53, 78, 6], "d": [31, 54, 87, 32], "e": [34, 56, 33, 67]})

print(df1)
print(df2)
```

	a	b
0	34	32
1	56	67
2	45	96
3	67	33

	c	d	e
0	89	31	34
1	53	54	56
2	78	87	33
3	6	32	67

In [5]:

```
print(df1.append(df2))
```

	a	b	c	d	e
0	34.0	32.0	NaN	NaN	NaN
1	56.0	67.0	NaN	NaN	NaN
2	45.0	96.0	NaN	NaN	NaN
3	67.0	33.0	NaN	NaN	NaN
0	NaN	NaN	89.0	31.0	34.0
1	NaN	NaN	53.0	54.0	56.0
2	NaN	NaN	78.0	87.0	33.0
3	NaN	NaN	6.0	32.0	67.0

In [19]:

```
info = pd.DataFrame([[4,9],[16,25],[36,49]],columns = ['x','y'])
print(info)
```

	x	y
0	4	9
1	16	25
2	36	49

In [20]:

```
import numpy as np
info.apply(np.sqrt)
```

Out[20]:

	x	y
0	2.0	3.0
1	4.0	5.0
2	6.0	7.0

In [21]:

```
info.apply(np.sum, axis=0) #column wise addition
```

Out[21]:

```
x    56
y    83
dtype: int64
```

In [22]:

```
info.apply(np.sum, axis=1) #row wise addition
```

Out[22]:

```
0    13
1    41
2    85
dtype: int64
```

In [28]:

```
print(info)
print(info.agg(['sum', 'min', 'max'])) )

print(info.agg(['sum', 'min', 'max'], axis = 0) )
print(info.agg(['sum', 'min', 'max'], axis = 1) )
```

```
   x  y
0   4  9
1  16 25
2  36 49

      x  y
sum  56  83
min   4   9
max  36  49

      x  y
sum  56  83
min   4   9
max  36  49

      sum  min  max
0      13    4    9
1      41   16   25
2      85   36   49
```

In [35]:

```
print(type(info.agg({'x' : ['sum', 'min', 'max'], 'y' : ['min', 'max', 'sum']})))
info.agg({'x' : ['sum', 'min', 'max'], 'y' : ['min', 'max', 'sum']})
```

```
<class 'pandas.core.frame.DataFrame'>
```

Out[35]:

	x	y
max	36	49
min	4	9
sum	56	83

In [36]:

```
print(info)
```

	x	y
0	4	9
1	16	25
2	36	49

In [51]:

```
print(type(info))
info = info.assign(Name = ['Smith', 'Parker', 'John'], age = [45,23,np.nan])
print(info)
```

```
<class 'pandas.core.frame.DataFrame'>
```

	x	y	Name	age
0	4	9	Smith	45.0
1	16	25	Parker	23.0
2	36	49	John	NaN

In [52]:

```
print(info.count())
```

```
x      3
y      3
Name    3
age     2
dtype: int64
```

In [53]:

```
print(info.count(axis = 1))
```

```
0      4
1      4
2      3
dtype: int64
```

In [54]:

```
import pandas as pd
import numpy as np

info=pd.DataFrame(np.random.randn(10,2),index=[1,3,7,2,4,5,9,8,0,6],columns=['col2', 'col1'])
print(info)
```

	col2	col1
1	0.475768	-0.216846
3	-0.692836	0.418266
7	0.132081	0.601276
2	-0.532736	0.003174
4	0.516909	-0.363466
5	-1.135711	0.953724
9	2.005558	0.417753
8	-1.959542	-1.143571
0	0.723175	1.538405
6	0.866467	-1.256980

In [55]:

```
info2 = info.sort_index()
print(info2)
```

	col2	col1
0	0.723175	1.538405
1	0.475768	-0.216846
2	-0.532736	0.003174
3	-0.692836	0.418266
4	0.516909	-0.363466
5	-1.135711	0.953724
6	0.866467	-1.256980
7	0.132081	0.601276
8	-1.959542	-1.143571
9	2.005558	0.417753

In [56]:

```
info3 = info.sort_index(ascending = False)
print(info3)
```

	col2	col1
9	2.005558	0.417753
8	-1.959542	-1.143571
7	0.132081	0.601276
6	0.866467	-1.256980
5	-1.135711	0.953724
4	0.516909	-0.363466
3	-0.692836	0.418266
2	-0.532736	0.003174
1	0.475768	-0.216846
0	0.723175	1.538405

In [59]:

```
info4 = info.sort_index(axis = 1)
print(info4)
```

	col1	col2
1	-0.216846	0.475768
3	0.418266	-0.692836
7	0.601276	0.132081
2	0.003174	-0.532736
4	-0.363466	0.516909
5	0.953724	-1.135711
9	0.417753	2.005558
8	-1.143571	-1.959542
0	1.538405	0.723175
6	-1.256980	0.866467

In [62]:

```
import pandas as pd
import numpy as np
info = pd.DataFrame({'col1':[7,1,8,3], 'col2':[8,12,4,9]})
print(info)
info_2 = info.sort_values(by='col2')
print(info_2)

info_3 = info.sort_values(by='col1')
print(info_3)
```

	col1	col2
0	7	8
1	1	12
2	8	4
3	3	9

	col1	col2
2	8	4
0	7	8
3	3	9
1	1	12

	col1	col2
1	1	12
3	3	9
0	7	8
2	8	4

In [64]:

```
import pandas
df = pandas.read_csv('G:\\Python Session\\python.csv')
print(df)
```

	name	department	Age
0	Sahil	Accounting	45
1	Joe	IT	54
2	Rohan	CSE	34
3	Sachit	HR	23

In [66]:

```
import pandas as pd
writer = pd.ExcelWriter('G:\\Python Session\\output.xlsx')
df.to_excel(writer)
writer.save()
print('DataFrame is written successfully to the Excel File.')
```

DataFrame is written successfully to the Excel File.

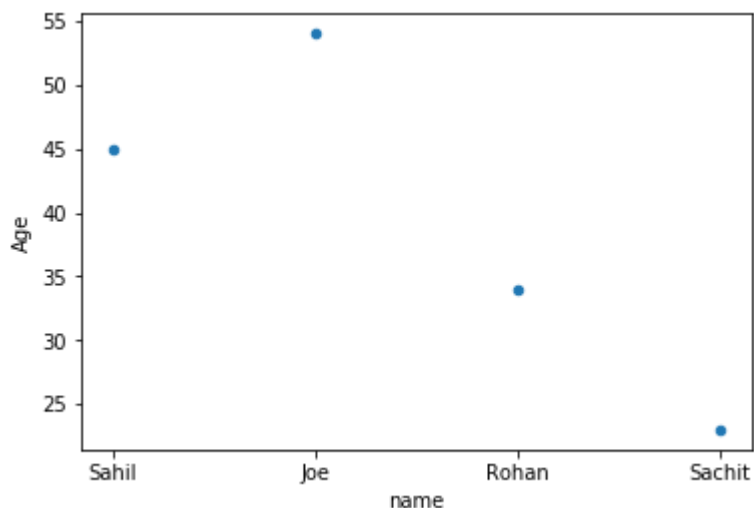
In [67]:

```
print(df)
```

	name	department	Age
0	Sahil	Accounting	45
1	Joe	IT	54
2	Rohan	CSE	34
3	Sachit	HR	23

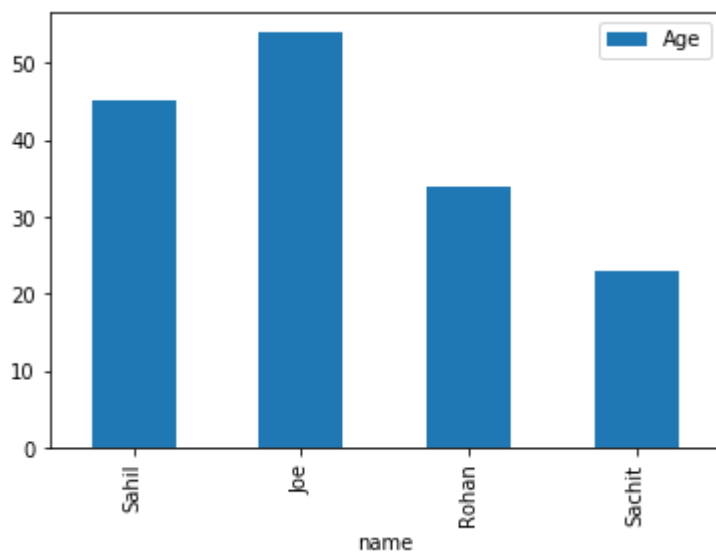
In [69]:

```
import matplotlib.pyplot as plt
df.plot(kind = 'scatter', x = 'name', y = 'Age')
plt.show()
```



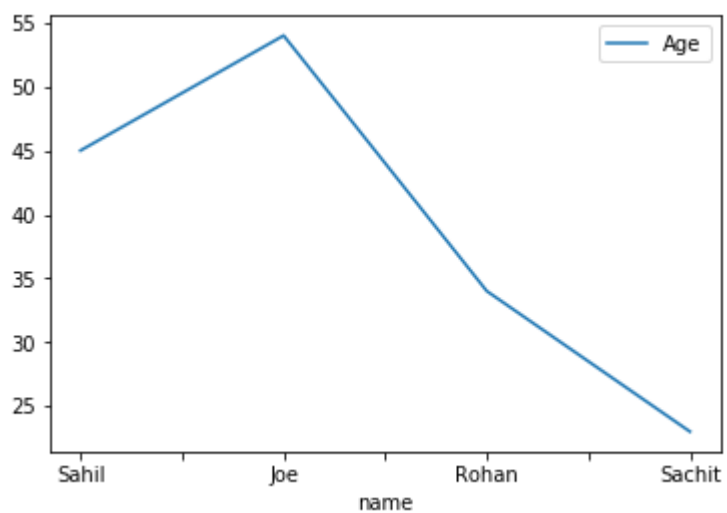
In [70]:

```
df.plot(kind = 'bar', x = 'name', y = 'Age')  
plt.show()
```



In [71]:

```
df.plot(kind = 'line', x = 'name', y = 'Age')  
plt.show()
```



In [79]:

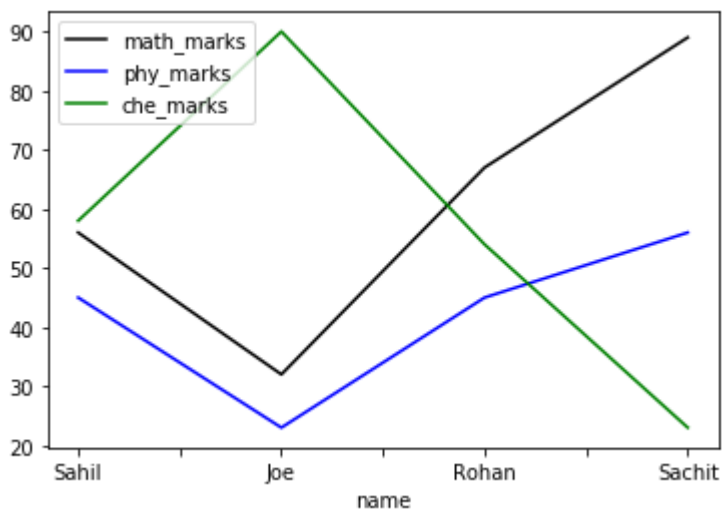
```
print(df)
df = df.assign(math_marks = [56,32,67,89], phy_marks = [45,23,45,56], che_marks = [58,90,54,23])
print(df)
```

	name	department	Age	math_marks	phy_marks	che_marks
0	Sahil	Accounting	45	56	45	58
1	Joe	IT	54	32	23	90
2	Rohan	CSE	34	67	45	54
3	Sachit	HR	23	89	56	23

	name	department	Age	math_marks	phy_marks	che_marks
0	Sahil	Accounting	45	56	45	58
1	Joe	IT	54	32	23	90
2	Rohan	CSE	34	67	45	54
3	Sachit	HR	23	89	56	23

In [80]:

```
import matplotlib.pyplot as plt
ax = plt.gca()
df.plot(kind = 'line', x = 'name', y = 'math_marks', color = 'black', ax=ax)
df.plot(kind = 'line', x = 'name', y = 'phy_marks', color = 'blue', ax=ax)
df.plot(kind = 'line', x = 'name', y = 'che_marks', color = 'green', ax=ax)
plt.show()
```

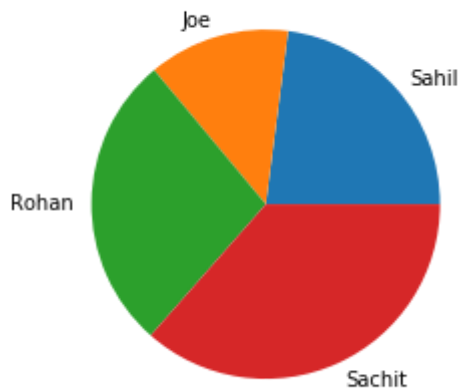


In [95]:

```
import matplotlib.pyplot as plt
print(df)
plt.title("Pie chart for Math marks")
plt.pie(df['math_marks'], labels = list(df['name']))
plt.show()
```

	name	department	Age	math_marks	phy_marks	che_marks
0	Sahil	Accounting	45	56	45	58
1	Joe	IT	54	32	23	90
2	Rohan	CSE	34	67	45	54
3	Sachit	HR	23	89	56	23

Pie chart for Math marks

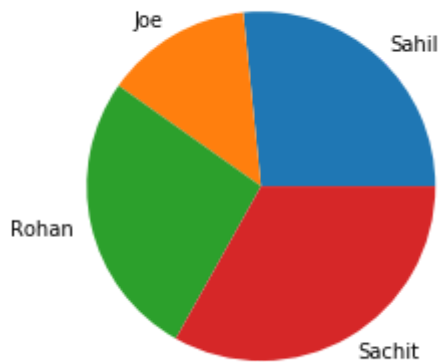


In [96]:

```
import matplotlib.pyplot as plt
print(df)
plt.title("Pie chart for phy marks")
plt.pie(df['phy_marks'], labels = list(df['name']))
plt.show()
```

	name	department	Age	math_marks	phy_marks	che_marks
0	Sahil	Accounting	45	56	45	58
1	Joe	IT	54	32	23	90
2	Rohan	CSE	34	67	45	54
3	Sachit	HR	23	89	56	23

Pie chart for phy marks

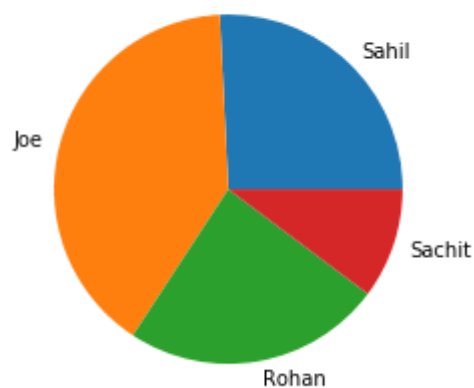


In [97]:

```
import matplotlib.pyplot as plt
print(df)
plt.title("Pie chart for Che marks")
plt.pie(df['che_marks'], labels = list(df['name']))
plt.show()
```

	name	department	Age	math_marks	phy_marks	che_marks
0	Sahil	Accounting	45	56	45	58
1	Joe	IT	54	32	23	90
2	Rohan	CSE	34	67	45	54
3	Sachit	HR	23	89	56	23

Pie chart for Che marks



In []: